



Figure 8: Interpolations of CelebA-HQ 256x256 images with 500 timesteps of diffusion.

be a fully expressive conditional distribution. With these choices, $D_{\text{KL}}(q(\mathbf{x}_T) \parallel p(\mathbf{x}_T)) = 0$, and minimizing $D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))$ trains p_θ to copy coordinates $t+1, \dots, T$ unchanged and to predict the t^{th} coordinate given $t+1, \dots, T$. Thus, training p_θ with this particular diffusion is training an autoregressive model.

We can therefore interpret the Gaussian diffusion model (2) as a kind of autoregressive model with a generalized bit ordering that cannot be expressed by reordering data coordinates. Prior work has shown that such reorderings introduce inductive biases that have an impact on sample quality [38], so we speculate that the Gaussian diffusion serves a similar purpose, perhaps to greater effect since Gaussian noise might be more natural to add to images compared to masking noise. Moreover, the Gaussian diffusion length is not restricted to equal the data dimension; for instance, we use $T = 1000$, which is less than the dimension of the $32 \times 32 \times 3$ or $256 \times 256 \times 3$ images in our experiments. Gaussian diffusions can be made shorter for fast sampling or longer for model expressiveness.

4.4 Interpolation

We can interpolate source images $\mathbf{x}_0, \mathbf{x}'_0 \sim q(\mathbf{x}_0)$ in latent space using q as a stochastic encoder, $\mathbf{x}_t, \mathbf{x}'_t \sim q(\mathbf{x}_t|\mathbf{x}_0)$, then decoding the linearly interpolated latent $\bar{\mathbf{x}}_t = (1 - \lambda)\mathbf{x}_0 + \lambda\mathbf{x}'_0$ into image space by the reverse process, $\bar{\mathbf{x}}_0 \sim p(\mathbf{x}_0|\bar{\mathbf{x}}_t)$. In effect, we use the reverse process to remove artifacts from linearly interpolating corrupted versions of the source images, as depicted in Fig. 8 (left). We fixed the noise for different values of λ so $\mathbf{x}_t$ and $\mathbf{x}'_t$ remain the same. Fig. 8 (right) shows interpolations and reconstructions of original CelebA-HQ 256×256 images ($t = 500$). The reverse process produces high-quality reconstructions, and plausible interpolations that smoothly vary attributes such as pose, skin tone, hairstyle, expression and background, but not eyewear. Larger t results in coarser and more varied interpolations, with novel samples at $t = 1000$ (Appendix Fig. 9).

5 Related Work

While diffusion models might resemble flows [9, 46, 10, 32, 5, 16, 23] and VAEs [33, 47, 37], diffusion models are designed so that q has no parameters and the top-level latent $\mathbf{x}_T$ has nearly zero mutual information with the data $\mathbf{x}_0$. Our ϵ -prediction reverse process parameterization establishes a connection between diffusion models and denoising score matching over multiple noise levels with annealed Langevin dynamics for sampling [55, 56]. Diffusion models, however, admit straightforward log likelihood evaluation, and the training procedure explicitly trains the Langevin dynamics sampler using variational inference (see Appendix C for details). The connection also has the reverse implication that a certain weighted form of denoising score matching is the same as variational inference to train a Langevin-like sampler. Other methods for learning transition operators of Markov chains include infusion training [2], variational walkback [15], generative stochastic networks [1], and others [50, 54, 36, 42, 35, 65].

By the known connection between score matching and energy-based modeling, our work could have implications for other recent work on energy-based models [67–69, 12, 70, 13, 11, 41, 17, 8]. Our rate-distortion curves are computed over time in one evaluation of the variational bound, reminiscent of how rate-distortion curves can be computed over distortion penalties in one run of annealed importance sampling [24]. Our progressive decoding argument can be seen in convolutional DRAW and related models [18, 40] and may also lead to more general designs for subscale orderings or sampling strategies for autoregressive models [38, 64].